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Article

Decanoic Acid Based Deep Eutectic Solvents: Constituent-Controlled Physicochemical Properties and Solvatochromic Probe Behavior

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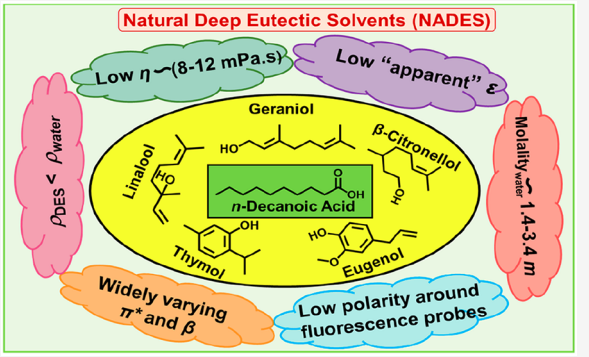
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ABSTRACT: The search for environmentally benign sustainable replacements for volatile organic compounds (VOCs) has resulted in advent of deep eutectic solvents (DESs) that have exhibited enormous application potential. Apart from the fact that their constituents could be inexpensive and naturally occurring, the physicochemical properties of the DESs can be easily tailored by judicious and systematic alteration of their constituents. Physicochemical properties and solute solvation environment afforded by five so-called "hydrophobic" natural DESs (NADESs) prepared by mixing *n*-decanoic acid (DA) with terpenoids thymol (Thy), geraniol (Ger), linalool (Lin), eugenol (Eug), and *β*-citronellol (Cit), respectively, in 1:1 mol ratio are investigated. Significantly low water miscibility in the DA-based DESs renders them viable nonaqueous media in lieu of currently used organic solvents. Their densities are lower than that of water and higher than most of the common "hydrophilic" DESs. The refractive index of the DESs scale with the density. The dynamic viscosity of these DESs are relatively lower as compared to the common tetralkylammonium salt- and metal-salt-based DESs thus enhancing their utility due to faster diffusion and dynamics. Their surface tension values are similar to those of several common organic solvents but lower than those of common "hydrophilic" DESs. Kamlet−Taft parameters, obtained using UV−vis absorbance probes, reveal the dipolarity/polarizability (π^*) to be much higher for phenyl group containing DESs (DA with Eug and Thy) in comparison to the others, whereas H-bond-donating acidity (α) does not vary much for the DESs. The H-bond-accepting basicity (β) spans wide range depending on the position of the hydroxyl group on the constituents. The solvation milieu of the ratiometric fluorescence probe pyrene is more polar for the phenyl-containing DESs as compared to the others. Five different fluorescence wavelength-shift solvatochromic probes reveal the solvation environment afforded by these DA-based DESs to be considerably nonpolar in nature. The outcomes reported here establish the DA-based NADESs as possible replacements for nonpolar organic solvents.



INTRODUCTION

Deep eutectic solvents (DESs) have recently developed as relatively benign and fairly low-priced solvent media with the possibility of widely differing physicochemical properties.[1−5] Subsequently, DESs have been extensively employed as solvent milieu in several applications ranging from catalysis, electroplating, solvation, drug solubilization, analytical chemistry, material science, and CO_2 capture among others.[6−11] Usually, the DESs are prepared by mixing two or more constituents [one or more constituent(s) may be solid under ambient conditions] in predetermined mole ratio(s) followed by gentle heating in many cases to afford liquid-state substances under ambient conditions. For several classes of DESs, this liquid-state under ambient conditions is a consequence of extensive H-bonding prevailing within the system. The extent of H-bonding and other interactions (electrostatic, van der Waals, etc.) present within a DES subsequently control the solvent−solvent and solute−solvent interactions afforded by the DES. While solvent−solvent (i.e., inter- and intraconstituent) interactions within DESs are investigated using plethora of spectroscopic techniques along with applying various computational/theoretical approaches, solute−solvent interactions within DESs are needed to be investigated to establish these solvents as routine solubilizing milieu for widespread applications in chemical and biological sciences.

The physicochemical properties of DESs are strongly contingent on their constituents; for a DES, the relative amounts of the constituents control the properties of that DES. Consequently, depending on the identity of the constituents and their relative amounts, DESs offer widely varying physicochemical properties. For example, depending on the constituents, a DES may be classified as "hydrophobic"

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affording very low water miscibility or "hydrophilic" having complete water miscibility. Even within the classification of "hydrophobic" or "hydrophilic", several physicochemical properties of a DES may still vary considerably depending on the identity of the constituents and their relative amounts. While physical properties, for instance, density, viscosity, refractive index, and surface tension, are continuously being reported for a variety of DESs in the current literature,[12−16] information derived from solute solvation, such as dipolarity/polarizability and H-bond donating (HBD) acidity/H-bond accepting (HBA) basicity as sensed by various spectroscopic probes, are still scarce. This dearth of information is imposing limitations on the applications of DESs.

In this work, we have employed a battery of spectroscopic probes to attain a thorough and conclusive understanding of the variation in solute−solvent interactions within DESs arising due to the differences in the identity of the constituents within a class of DESs. Several common UV−vis absorbance and fluorescence-based probes are employed, and based on their optical spectroscopic responses, corresponding empirical parameters are estimated.[17−19] Along with the solvatochromic probe responses, essential physical properties including density, dynamic viscosity, surface tension, and refractive index are also measured. A wide array of DESs composed of *n*-decanoic acid (DA), a common and popular long chain fatty acid, combined with five (5) different terpenoids (thymol (Thy), geraniol (Ger), linalool (Lin), eugenol (Eug), and *β*-citronellol (Cit)) are investigated. These terpenes-based DESs are often referred to as natural deep eutectic solvents (NADES).[20] The abundance of these natural resources accompanied by their high hydrophobicity and low viscosity has rendered them exceedingly advantageous for widespread use in drug delivery, biomass extraction, chemical reactions, separation, and processing, among others.[21−30] Moreover, NADESs are relatively novel and underutilized. Hence, it is a scientific requisite to explore physicochemical properties as well as solute solvation within nascent NADESs. The unified approach in this research, collating evidence collected from various structurally varying fluorophores and absorbance probes exercised in a series of optical spectroscopic methodologies, affords key molecular-level understanding of solute−solvent (i.e., probe−DES) interactions. All things considered, our multiprobe approach provides a conclusive and cohesive view of solvent interactions within common and popular DESs indicating a promising path forward for exploring DESs, in general.

EXPERIMENTAL SECTION

Materials. DES constituents, *n*-decanoic acid (DA, ≥99%), thymol (Thy, ≥99%), eugenol (Eug, ≥99%), *β*-citronellol (Cit, ≥95%), geraniol (Ger, ≥98%), and linalool (Lin, ≥97%), were acquired from Sigma-Aldrich and were employed without further processing. HPLC-grade water with highest purity was purchased from Thermo Fisher Scientific India Pvt. Ltd. and was used without further processing. Structures of the DES constituents are presented in Figure 1. In order to prepare DESs, DA and terpenoids were mixed well in 1:1 mol ratio and the mixture was continuously stirred at 50 °C to get homogeneous and clear solutions.

N,*N*-Diethyl-4-nitroaniline (DENA) and 4-nitroaniline (NA) were procured from Frinton Laboratories and Spectrochem Co. Ltd., respectively, and Nile Red [NR, ≥98% (HPLC)] was acquired from Sigma-Aldrich and was utilized as collected. The structures of the absorbance probes used in the work are presented in Figure 2A. 1-Anilino-8-napthlene sulfonate (ANS, 99%), *p*-toluidinyl-6-napthalenesulfonate (TNS, 99%), and 6-propionyl-2-(dimethylaminonaphthalene) [PRODAN, ≥98% (HPLC)] were used as fluorescence probes and were bought from Biochemika. Other florophore, pyrene [Py, ≥99% (GC)], 1-pyrenecarboxaldehyde (PyCHO, 99%), and coumarin-153 (C153, dye content 99%) were acquired from Sigma-Aldrich and were exercised without further processing. The structures of the investigated fluorescence probes are shown in Figure 2B.

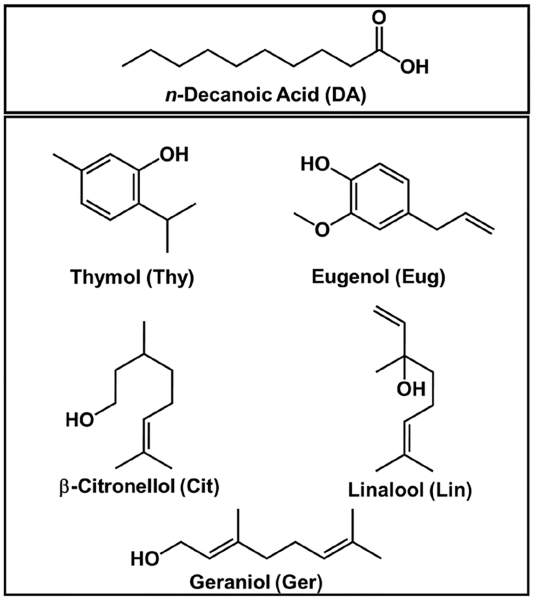


Figure 1. Constituents of the DA-based DESs investigated in this work.

Methods. Water miscibility was determined using visual method by gradually adding HPLC-grade water to the DES. A Mettler-Toledo delta-range density meter (DE45) was effectively employed to determine the densities (ρ) of the freshly prepared DESs. The density measuring principle of this instrument is founded on the electromagnetically induced oscillations of the sample filled U-shaped glass tube. The density measurements of all DESs were made at 298.15 K and atmospheric pressure. The calibration was conducted utilizing air and double-distilled water. The measurement of dynamic viscosities (η) of the freshly prepared DESs was performed by means of a Peltier-based Anton Paar microviscometer containing precalibrated glass capillaries of varying dimensions (0.01 K resolution and < ±0.05 K accuracy). The instrument function on the basis of the rolling-ball principle, which follows the movement of a ball down the slanting sample-filled capillaries. The calibration was performed by means of double-distilled water. The refractive index (n_D) of each investigated DES was quantified using a digital ABBE refractometer (Labtronics, LT-738) with accuracy of ±0.0002. The calibration of the ABBE refractometer instrument was accomplished by comparing the consistency of the measured refractive indices of acetone and distilled water at ambient temperature (298.15 K) and the reported values in literature. A model DCAT 11EC tensiometer with Wihelmy plate procured from Data Physics was employed to determine the surface tension (γ) of the DA-based DESs. The platinum plate was first rinsed with ethanol before every measurement to remove

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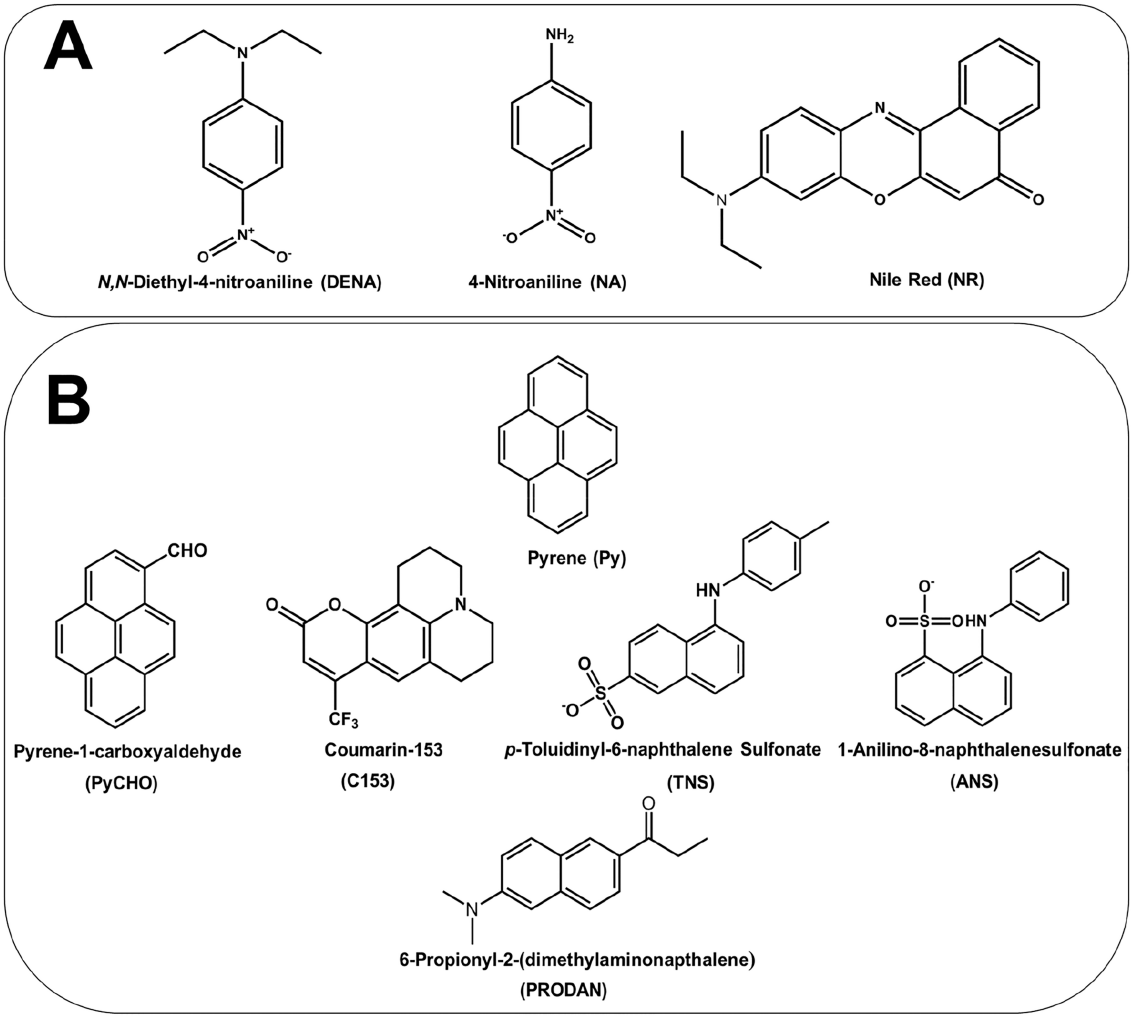


Figure 2. Molecular structures of the absorbance probes (panel A) used to estimate Kamlet−Traft parameters and solvatochromic fluorescence probes (panel B) used in this work.

Table 1. Miscibility of Water within DA-Based DESs at 298.15 K at Ambient Pressure

DES	(*v*/*v*)%	(*w*/*w*)%	(*v*/*w*)%	molarity (M)	molality (*m*)
DA:Eug	$5.6 \pm 0.1_5$	$5.7 \pm 0.1_5$	$5.7 \pm 0.1_5$	$3.1 \pm 0.1_3$	$3.4 \pm 0.1_5$
DA:Thy	$3.9 \pm 0.1_4$	$4.2 \pm 0.1_4$	$4.2 \pm 0.1_4$	$2.2 \pm 0.1_2$	$2.4 \pm 0.1_3$
DA:Ger	$2.2 \pm 0.1_1$	$2.5 \pm 0.1_1$	$2.5 \pm 0.1_2$	$1.2 \pm 0.1_0$	$1.4 \pm 0.1_0$
DA:Lin	$3.4 \pm 0.1_3$	$3.8 \pm 0.1_3$	$3.8 \pm 0.1_5$	$1.9 \pm 0.1_1$	$2.2 \pm 0.1_3$
DA:Cit	$4.1 \pm 0.1_4$	$4.6 \pm 0.1_4$	$4.7 \pm 0.1_4$	$2.3 \pm 0.1_2$	$2.7 \pm 0.1_4$

organics materials sticking to the plate, and finally heating it to red hot with a Bunsen burner.

Stock solutions for spectroscopic studies were prepared by dissolving precalculated amount of each probe in ethanol and stored at 4 ± 1 °C in precleaned amber glass vials. Mettler-Toledo AB104-S balance was used to weigh the required amount of probe with accuracy of ±0.0001 g. A prerequisite amount of the prepared stock solution was taken in a 1 cm^2 quartz cuvette. Grade 1 nitrogen gas was used to evaporate the ethanol. The required amount of the DES under investigation was transferred to the cuvette. Dissolution of the probes in DESs were confirmed using the linearity of the absorbance versus the concentration plot. The UV−vis absorbance spectra of all the samples were obtained using a Lambda 35 double-beam spectrophotometer (purchased from PerkinElmer) with an adjustable bandwidth and containing deuterium and a Xe lamp as light sources and a photomultiplier tube (PMT) detector. The steady-state fluorescence spectra were attained on a FLS1000 spectrometer with emission monochromators (STGM325-M) and grating excitation (STGM325-X) containing a 450 W xenon arc lamp as the radiation source, a red PMT detector, and a temperature-controller (a Quantum Northwest Luma 40). The spectrometer was procured from Edinburgh Instruments, Ltd. Appropriate blank was subtracted from the emission and excitation spectra before analysis. Data analysis was accomplished using SigmaPlot v14.5.

RESULTS AND DISCUSSION

Miscibility of Water in DA-Based DESs. Assessment of water intake by a DES is of utmost importance. In many usages and applications of a solvent, it is important to either remove

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residual water, add external water, or estimate the amount of water present. Water miscibility within the five DA-based DESs under ambient conditions are assessed and reported in Table 1. The water intake varies between 1.4 *m* and 3.4 *m* within the five DESs and follows the trend DA:Eug > DA:Cit > DA:Thy > DA:Lin > DA:Ger. While most choline chloride based DESs exhibit complete water miscibility, the water miscibility within these DA-based DESs is significantly lower than those in tetraalkylammonium salt containing DA-based DESs.[31] The relatively low water intake by these DESs renders them almost "hydrophobic" in nature and open further avenues of their applications as novel nonaqueous media.

Density, Refractive Index, Dynamic Viscosity, and Surface Tension of DA-Based DESs. Density (ρ), refractive index (n_D), dynamic viscosity (η), and surface tension (γ) of the five DA-based DESs under ambient conditions are listed in Table 2. A thorough examination of the density values shows

Table 2. Density (ρ), Refractive Index (n_D), Dynamic Viscosity (η), and Surface Tension (γ) of the DA-Based DESs at 298.15 K at Ambient Pressure[a]

DES	ρ (g·cm^{-3})	n_D	η (mPa·s)	γ (mN m^{-1})
DA:Eug	0.970	1.4767	8.20	30.67
DA:Thy	0.930	1.4679	11.83	29.66
DA:Ger	0.887	1.4494	8.62	29.04
DA:Lin	0.879	1.4416	7.84	27.69
DA:Cit	0.876	1.4402	9.75	28.43

[a]Errors (in terms of ±1 standard deviation) in ρ, n_D, η, and γ measurements are ±0.002 g·cm^{-3}, ±0.0005, ±0.02 mPa·s, and ±0.05 mN·m^{-1}, respectively.

that irrespective of the identity of the constituent, the densities of all the investigated DESs are lower than that of water under ambient conditions. Furthermore, the densities of these "hydrophobic" DA-based DESs are evidently lower than most of the common "hydrophilic" DESs (metal salt, hydrated metal salt, choline salt, and tetraalkylammonium salt based DESs, among others), many ionic liquids (ILs), and low-molecular-weight polyethylene glycols.[32−35] It is fascinating to note that the densities of the five investigated DESs are not very different from one another (the highest density of 0.970 g·cm^{-3} and the lowest 0.876 g·cm^{-3} for DA:Eug and DA:Cit, respectively). The observed constituent-dependent density variation of these DA-based DESs is easily attributed to the density of each of the constituent of the DA-based DES − the density of the five DESs exhibits the same trend as the density of the constituents (at 298.15 K, the density values of Eug, Thy, Ger, Lin, and Cit are 1.065, 0.970, 0.889, 0.857, and 0.855 g·cm^{-3}, respectively).[36,37]

The refractive index (n_D) of the five DA-based DESs varies between 1.4767 and 1.4402 and follows the same trend as that observed for density (Table 2). It is proposed, in general, that the higher the density of a medium (i.e., the more molecules/ions present per unit volume), the more the light is being obstructed resulting in increase in refractive index of the medium. The constitution of matter and mechanism of the propagation of light results in different formulas for the dependence of the refractive index on density that imply that higher the density of the medium the higher the refractive index.[38] The refractive index of DA:Eug was found to be similar to that of a tetrabutylammonium bromide:3-amino-1-propanol (TBAB:AP) DES in 1:6 molar ratio (1.4766), while the refractive index values of the remaining four DA-based DESs are lower than the corresponding values for tetraalkylammonium salt-based DESs.[35]

The measured dynamic viscosity (η) of the five DA-based DESs lies between 11.83 and 7.84 mPa·s and are reported in Table 2. A comparison of the viscosity values of the DESs investigated here with the reported viscosities of some of the other DESs reveals the viscosities the DA-based DESs to be relatively lower. For example, viscosity is much higher for type III DESs prepared from choline chloride and urea eutectics (ChCl:urea) in 1:2 molar ratio (750 mPa·s at 298.15 K)[39] and similarly for type I DES consisting of $ZnCl_2$ and urea eutectic mixture (11340 mPa·s at 298.15 K) in 1:3.5 molar ratio.[40] The viscosities of hydrated metal salt: urea DESs (La:urea with 136.9124 mPa·s, Ce:urea with 123.7911 mPa·s, and Gd:urea with 136.3322 mPa·s; all at 298.15 K and in 1:5 molar ratio) are also higher as compared to the viscosities of the DA-based DESs investigated here.[41] The relatively lower viscosity of these DESs imparts them additional advantages as diffusion and dynamics is supposed be faster in media with relatively lower viscosities.

Surface tension (γ) values of the five DA-based DESs are found to vary between 30.67 mN·m^{-1} (for DA:Eug) to 27.69 mN·m^{-1} (for DA:Lin) (Table 2). At 298.15 K, the other common solvents having surface tension within this range are as follows: acetonitrile (28.66 mN·m^{-1}), benzene (28.21 mN·m^{-1}), 1-decanol (28.02 mN·m^{-1}), octadecane (27.87 mN·m^{-1}), toluene (27.91 mN·m^{-1}), and xylenes (29.51, 28.12, and 27.80 mN·m^{-1} for *o*-, *m*-, and *p*-, respectively).[37] Similar to the other physical properties, the surface tension of these NADESs are also much lower as compared to the surface tension of the ChCl, methyltriphenylphosphonium bromide, and *N*,*N*-diethylenethanol ammonium chloride based DESs.[32]

Kamlet−Taft Parameters of the DA-Based DESs. Empirical Kamlet−Taft parameters, notably dipolarity/polarizability (π^*), H-bond donating (HBD) acidity (α), and H-bond accepting (HBA) basicity (β), are used extensively to establish physicochemical properties of a solubilizing media with reference to solutes.[42−44] As reported earlier, responses from suitable solvatochromic absorbance probes (Nile red (NR),[44] *N*,*N*-diethyl-4-nitroaniline (DENA)[42] and 4-nitroaniline (NA))[43] are used to estimate parameters π^*, α, and β for the five DA-based DESs using the following expressions:

$$\pi^* = 8.649 - 0.314\bar{\upsilon}_{DENA} \quad (1)$$

$$\alpha = \frac{20.47 - 1.95\pi^* - \bar{\upsilon}_{NR}}{1.01} \quad (2)$$

$$\beta = -0.357\bar{\upsilon}_{NA} - 1.176\pi^* + 11.12 \quad (3)$$

where $\bar{\upsilon}_{DENA}$, $\bar{\upsilon}_{NR}$, and $\bar{\upsilon}_{NA}$ are the lowest energy absorbance band maxima of the probes DENA, NR, and NA, respectively (Figure S1), in kK units (1000 cm^{-1} = 1 kK). The parameters π^*, α, and β thus estimated are reported in Table 3. While π^* is found to be the highest for DA:Eug DES (π^* = 0.88) followed closely by DA:Thy DES (π^* = 0.86), for the other three DESs, the π^* are significantly lower in comparison (π^* = 0.56, 0.45, and 0.45 for DA:Ger, DA:Lin, and DA:Cit, respectively). Clearly, the presence of phenyl ring with conjugated π-electrons in Eug and Thy manifests itself in higher observed dipolarity/polarizability of DA:Eug and DA:Thy DESs, respectively; the presence of additional ether and alkene groups on Eug results in slightly higher π^* for the

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Table 3. Kamlet−Taft Parameters: Dipolarity/Polarizability (π^*), HBD Acidity (α), and HBA Basicity (β) of the DA-Based DESs at 298.15 K[a]

	Kamlet−Taft parameters		
DES	π^*	α	β
DA:Eug	0.88	0.77	0.94
DA:Thy	0.86	1.06	0.11
DA:Ger	0.56	0.99	0.71
DA:Lin	0.45	0.82	0.81
DA:Cit	0.45	1.06	0.73

[a]The standard error in π^*, α, and β are less than or equal to ±0.02.

DES constituted of DA and Eug. While one hydroxyl group is also present in Ger, Lin, and Cit along with the double bonds (two each in Ger and Lin, and one in Cit), the absence of phenyl group renders lower π^* for DA-based DESs with these constituents. It is to be noted that π^* of most of the ILs are usually higher than the DA-based DESs investigated here due to the fact that ILs comprise ions alone.[45] DESs, on the other hand, can have widely varying π^* based on their constituents and composition. While the π^* of DA-based DESs with Ger, Lin, and Cit are comparable to other medium chain organic acid based DESs with neutral constituents, it is surprising to note that π^* of DA:Eug and DA:Thy are more similar to the medium chain organic acid based DESs with salts, such as tetraalkylammonium chloride ($N_{xxxx}Cl$) and choline chloride (ChCl), as other constituents. Specifically, π^* of DA:Eug and DA:Thy (π^* of 0.88 and 0.86, respectively) are higher than the π^* of DA:ChCl (2:1) and DA:$N_{4444}Cl$ (2:1), reported as 0.42[46] and 0.69[46,47] (or 0.73),[48] respectively. It may be inferred that the presence of a phenyl group on the DES constituents in higher proportion (equimolar in our DESs) imparts significantly higher dipolarity/polarizability to the medium.

The HBD acidity (α) of our DA-based DESs with −OH-containing constituents do not vary as much as the π^*; the highest α is estimated for DA:Thy and DA:Cit (α = 1.06 for both) and the lowest for DA:Eug (α = 0.77). These α values are similar to those of methanol and medium-chain organic acids. In comparison, the α of our DA-based DESs are similar to the $N_{xxxx}Cl$ DESs with medium chain organic acids; they are somewhat higher than the ChCl DESs with the same organic acids.[46,47] It may be concluded that the phenyl −OH versus alkyl −OH functionalities present in constituents do not affect the HBD acidity of the DA-based DESs to significant extent. The HBA basicity (β) of our DA-based DESs, however, does vary widely; with the highest β = 0.94 observed for DA:Eug and the lowest β = 0.11 for DA:Thy. It is easy to comprehend that, for the constituents having −OH present at the end of the organic chain (i.e., in Ger and Cit), the β are similar (0.71 and 0.73, respectively); for DA-based DES with Lin where the −OH is present at the allylic position, the β is higher (0.81). For Eug DES, the presence of − O− results in a high value of β, whereas only a phenolic −OH on the constituent Thy manifests itself through the lowest β (only 0.11).

Response of Fluorescence Probes in DA-Based DESs. Due to their inherent sensitivity, the fluorescence-based approaches to investigate solute solvation in complex solubilizing media have garnered major importance.[49] In order to investigate possible interactions including solute−solvent and solvent−solvent in the context of solvent dipolarity within the DA-based DESs, we have employed six different fluorescence probes whose responses are multidimensional. With this methodology, we envision to explore probe molecular architecture dependent interactions within our DESs so that the outcomes may be generalized to solutes with similar functionalities.

Response of Ratiometric Probe Pyrene. Pyrene is known as one of the most extensively exercised neutral fluorescence probes and it comprises four fused benzene rings and has no other functional group (Figure 2B). The solvent polarity scale for pyrene is outlined as the I_I/I_{III} ratio of emission band intensities, where band I represents the $S_1(v = 0) \rightarrow S_0(v = 0)$ transition and band III corresponds to the $S_1(v = 0) \rightarrow S_0(v = 1)$ transition.[50] The I_I/I_{III} emission intensity ratio rises with increase in solvent polarity.[50] Table 4 shows measured pyrene I_I/I_{III} within four of the five DA-based DESs (the pyrene emission response could not be collected in DA:Eug DES due to its inherent color giving rise to significant background signal) under ambient conditions. Figure 3 presents the pyrene emission spectra in the DESs investigated. The nonpolar nature of the pyrene cybotactic region within the four DA-based DESs is clearly manifested as a comparison reveals the pyrene I_I/I_{III} to be significantly higher in ChCl:HBD DESs: The pyrene I_I/I_{III} within the four DA-based DESs lies between those of *n*-butanol (1.05) and cyclohexane (0.56) (Table S1).[51] Furthermore, as expected, among the four DESs, pyrene I_I/I_{III} for DA:Thy is the highest (due to the presence of phenolic moiety in Thy), whereas it has similar values in other three DESs with nonaromatic constituents.

Responses of Wavelength-Shift Probes. One such probe, 1-pyrenecarboxaldehyde (PyCHO), shows two types of closely spaced excited states, (n, π^*) and (π, π^*). In nonpolar solvents, the emission originates from n−π^* transition with highly structured spectra and fairly low fluorescence quantum

Table 4. Ratio of Band I to Band III Emission Intensities of Pyrene (I_I/I_{III}), Fluorescence Emission Maxima (λ^{em}_{max}) of PyCHO, C153, TNS, and ANS, Fluorescence Emission Maxima (λ^{em}_{max}), Fluorescence Excitation Maxima (λ^{ex}_{max}), and Stokes shifts of PRODAN in DA-based DESs at 298.15 K[a]

	pyrene	PyCHO	C153	TNS	ANS	PRODAN		
DES	I_I/I_{III}	λ^{em}_{max}/nm				λ^{em}_{max}/nm	λ^{ex}_{max}/nm	Stokes shift/cm^{-1}
DA:Eug	NA	437	520	432	NA	474	395	4219
DA:Thy	0.87	429	517	436	480	466	386	4448
DA:Ger	0.79	441	516	437	479	478	394	4460
DA:Lin	0.77	437	504	437	472	464	392	3958
DA:Cit	0.76	437	512	437	476	472	371	5768

[a]Error in λ^{em}_{max} and λ^{ex}_{max} are ≤ ±2 nm and in Stokes shift is ≤ ±100 cm^{-1}. NA (not acquired): Could not acquire due to colored appearance leading to significant inherent signal from this DES.

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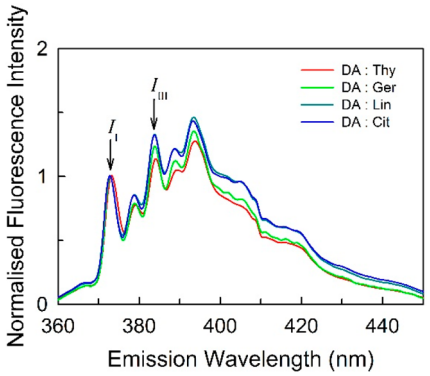


Figure 3. Fluorescence emission spectra of pyrene (25 μM, λ_{ex} = 337 nm) dissolved in DA-based DESs at 298.15 K.

yields (<0.001).[52] However, on increasing the polarity of the solubilizing media, the energy levels are inverted, i.e, the $\pi-\pi^*$ energy level becomes the lower energy level as compared to the $n-\pi^*$ level due to the solvent relaxation during its excited-state lifetime. Subsequently, now a broad, structureless, and moderately intense fluorescence emission originates from the $\pi-\pi^*$ transition. It has been evoked that a static dielectric constant (ε) of ~10 is essential to equalize the $n-\pi^*$ and $\pi-\pi^*$ energy levels.[52] More importantly, the emission maxima of PyCHO is known to show bathochromic shift with increasing ε of the surrounding milieu.[52] The fluorescence emission maxima (λ_{max}^{em}) for PyCHO in the five DA-based DESs are found to be in within 429−441 nm (Table 4, Figure S2). In comparison, the values are in the range of 452−460 nm for ChCl-based "hydrophilic" DESs (Table S1).[51] This implies the relative hydrophobic cybotactic region encountered by PyCHO when dissolved in DA-based DESs as opposed to ChCl-based DESs.

When another such wavelength-shift probe, coumarin-153 (C153), is electronically excited, the dipole moment value varies from 6.6 D in the ground-state to a value ranging from 14.2 to 16.0 D in the first excited-state S_1 state varying in accordance to the surrounding media.[53] The higher the solvent polarity, lower the relaxed-state energy, thus inducing a greater redshift in the emission spectra. The emission maxima of C153 in DA-based DESs are presented in Table 4 (Figure S2 presents the emission spectra). λ_{max}^{em} varies from 504 nm in DA:Lin to 520 nm in DA:Eug. In comparison, the values in ChCl-based "hydrophilic" DESs range between 540 and 524 nm,[51] implying these DA-based DESs to offer a relatively nonpolar solvation environment to C153. Long alkyl chain acid (i.e., DA) instead of ammonium salt, i.e., ChCl, as a constituent lead to substantially reduced cybotactic region dipolarity for C153 within the investigated DESs.

p-Toluidinylnaphthalene-6-sulfonate (TNS) is one of the frequently used fluorophore which fluoresces weakly in water but strongly in organic solvents.[54] It is established that TNS fluorescence is enhanced by solvents of low ε with an accompanying hypsochromic shift.[54] For example, the TNS emission maxima appears at ~413 nm in dioxane, whereas it is at ~500 nm in water.[54] In five different DA-based DESs, the TNS emission maxima does not vary much; it varies from 432 to 437 nm (Table 4 and Figure S2). This implies similar cybotactic region dipolarity around TNS irrespective of the constituents of the DA-based DESs. In comparison, the ChCl-based hydrophilic DESs have TNS emission maxima reported to be in the 455−473 nm range.[51]

Another similar probe, 8-anilinonaphthalene-1-sulfonate (ANS), which exhibits emission maxima at ~464 nm in ethanol that bathochromically shifts to ~523 nm in water, is utilized next.[55] Again, due to intense inherent emission from the DES itself, ANS response in DA:Eug DES could not be collected. In the other four DA-based DESs, the ANS emission maxima ranges between 472 and 480 nm (Table 4 and Figure S2). This again suggests no appreciable differences in the cybotactic region dipolarity as sensed by ANS. In comparison, in ChCl-based DESs with ethylene glycol, glycerol, and urea, the ANS emission maxima are reported to be 497, 492, and 490 nm, respectively.[51] These significantly higher values again establish low dipolarity solvation milieu offered by our DA-based DESs in comparison. It is convenient to note that the solvation environment afforded to both sulfonate-group probes with similar structures (TNS and ANS) by our DA-based DES are similar in nature irrespective of the structural variances in the DES constituents as well as in probe chemical structures. This observation allows the prediction of generalized solvation environment based on probe as well as constituent structure.

PRODAN Stokes Shift and Assessment of "Apparent" Static Dielectric Constant. One of the most popular and used twisted intramolecular charge transfer (TICT) probes is 6-propionyl-2-(dimethylamino)naphthalene (PRODAN).[56] While initially reported by Weber and Farris in 1979,[57] PRODAN has been employed extensively to probe various complex media. Catalan et al.,[58] introduced a new acidic term into the Lippert equation[49] for utilizing PRODAN as a polarity indicator for protein cavities predominantly if the examined environments are comprised of acidic sites, to satisfactorily describe the PRODAN absorption and emission spectroscopic behavior in several solvents. While PRODAN emission maxima in nonpolar solvents appear around 401 nm, it bathochromically shifts to around 531 nm in polar solvents.[56] In five DA-based DESs, the PRODAN emission maxima shows significant shift of 14 nm from 464 nm (in DA:Lin DES) to 478 nm (in DA:Ger DES) (Table 4, Figure S3). These emission maxima are at higher energies as compared to the emission maxima in ChCl-based DESs (499, 501, and 489 nm for DESs of ChCl:ethylene glycol, ChCl:glycerol, and ChCl:urea, respectively).[51] This again establishes relatively nonpolar solvation milieu offered to PRODAN by these DA-based DESs. Similar to the responses of neutral probes (pyrene, PyCHO, and C153) and contrary to those of ionic probes (TNS and ANS), the neutral probe PRODAN also exhibits constituent-dependent cybotactic region dipolarity within our DESs.

Stokes shifts [$(\overline{\upsilon}_A - \overline{\upsilon}_F)$, where $\overline{\upsilon}_A$ and $\overline{\upsilon}_F$ are the respective excitation and emission band maxima] of PRODAN versus the orientational polarizability ($\Delta f'$) for different milieu can be plotted following the empirical formulation suggested primarily by Lippert and Mataga (eqs 4 and 5).[49] The Lippert−Mataga equation is only an estimation with various approximation.

$$\overline{\upsilon}_A - \overline{\upsilon}_F = \frac{2}{hc}\left(\frac{\epsilon - 1}{2\epsilon + 1} - \frac{n^2 - 1}{2n^2 + 1}\right)\left(\frac{2n^2 + 1}{n^2 - 1}\right) \times \frac{(\mu_G^2 - \mu_E^2 - \mu_G\mu_E\cos\alpha)}{a^3} + \text{constant} \quad (4)$$

with

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$$\Delta f' = \left(\frac{\epsilon - 1}{2\epsilon + 1} - \frac{n^2 - 1}{2n^2 + 1}\right)\left(\frac{2n^2 + 1}{n^2 - 1}\right) \quad (5)$$

Though most of these approximation can be contemplated as valid in our case, a modified form that assumes the polarizability of the fluorophore to be the same as that of the solvent, and that μ_E and μ_G (probe dipole moments in the excited and ground states, respectively) point in the separate directions with the angle between them termed as α, eventually becomes more satisfactory in treating the solvatochromic behavior of probe in dissimilar solubilizing milieus (a is defined as the radius of the cavity where the fluorophore resides). Owing to the structural similarities, the majority of the restrictions can be dropped in our systems. Figure 4 exhibits

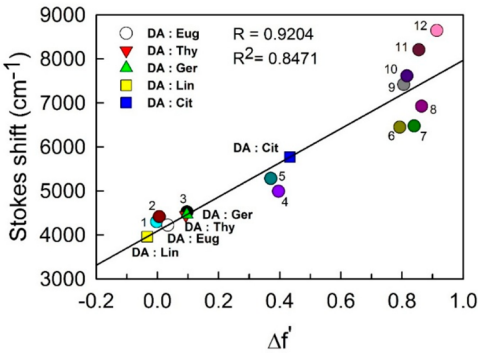


Figure 4. Lippert−Mataga plot of PRODAN dissolved in many solvents [1) cyclohexane, 2) benzene, 3) triethylamine, 4) chlorobenzene, 5) chloroform, 6) acetone, 7) *N*,*N*-dimethylformamide, 8) acetonitrile, 9) propylene glycol, 10) ethanol, 11) methanol, and 12) water] under ambient conditions. Stokes′ shifts of the five DA-based DESs are also encompassed.

the Stokes shifts versus $\Delta f'$ plots in various solvents. Regression analysis of the data unearths that the Stokes shifts and $\Delta f'$ shows satisfactory linear behavior for these types of empirical analyses (slope = 3881 ± 522 cm^{-1}, intercept = 4087 ± 345 cm^{-1}, and standard error of estimate = ±631).[51] From the experimentally determined excitation maxima of PRODAN combined with the emission maxima in Table 4, respective Stokes shifts were evaluated for the five DA-based DESs. The estimated Stokes shifts are further exploited to evaluate the $\Delta f'$ for the five DESs from the recovered parameters obtained from the linear regression analysis (Figure 4). Using our measured refractive index (n_D) values of the DA-based DESs (*vide supra*), "apparent" ε for the four DA-based DESs could be estimated. The "apparent" ε thus obtained are 3.1, 3.0, 2.5, and 1.9 for DA:Thy, DA:Ger, DA:Eug, and DA:Lin DESs, respectively. These "apparent" ε can be compared with those obtained earlier for ChCl-based DESs using the same strategy.[51]

The reported estimated values are 32 for ChCl:ethylene glycol DES, 22 for ChCl:glycerol DES, and 12 for ChCl:urea DES.[51] The considerably lower "apparent" ε of DA-based DESs clearly establish nonpolar nature of these DESs in the context of both ChCl-based DESs as well as common organic solvents in general.

CONCLUSIONS

The solubility of water in the five investigated DA-based DESs is between 2.5−5.7 wt % under ambient conditions. In comparison, solubility of water in ethyl acetate is ca. 3.3% and in methyl *tert*-butyl ether is ca. 1.5%. This puts hydrophobicity associated to these DA-based DESs in perspective. The density of these DESs is lower than that of water, and the refractive index follows the trend observed in density. While the surface tension of the DA family DESs are similar to several common organic solvents, the favorable low viscosity may render them solubilizing media of choice for chemical reactions, separation, extraction, and many other applications where faster diffusion and dynamics are required. Empirical dipolarity/polarizability and HBA basicity parameters obtained from UV−vis absorbance from solvatochromic probes do exhibit major dependence on chemical structure of the constituents of the DES: Constituents with a phenyl group impart higher dipolarity/polarizability to corresponding DESs and the location of hydroxyl group on a constituent controls the HBA basicity of the DES. All the investigated fluorescence polarity probes experience solvation environment of relatively low polarity (similar to that usually offered by nonpolar organic solvents) within the DA-based DESs as compared to common salt-based DESs. While the responses from neutral fluorescence probes are contingent on the constituent of the DES, the ionic fluorescence probes exhibit similar responses irrespective of the DES constituents possibly due to preferential solvation of such probes by DA as opposed to the terpenoids. Relatively low dipolarity of the solute cybotactic region combined with the hydrophobic nature of these DA-based DESs propels them as suitable environmentally benign and sustainable solvents to replace hazardous nonpolar organic solvents.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.iecr.2c02608.

Summary of reported ratio of band I-to band III emission intensities of pyrene Py (I_I/I_{III}) and fluorescence emission maxima (λ_{max}^{em}) of PyCHO, ANS, TNS, PRODAN, Coumarin-153 in various solvents; absorbance spectra of DENA, NA and NR dissolved in DA based DESs at 298.15 K; fluorescence emission spectra of PyCHO, C153, TNS, and ANS dissolved in DA-based DESs at 298.15 K; fluorescence emission spectra of PRODAN dissolved in DA-based DESs at 298.15 K (PDF)

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Notes

The authors declare no competing financial interest.

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